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Using Explainable Artificial Intelligence to Predict Internet Addiction in Turkish University Students

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ABSTRACT
The rising incidence of internet addiction (IA) among university students has raised concerns about its impact on academic performance, mental health, and social relationships. This study aims to model internet addiction in 1,180 university students using Explainable Artificial Intelligence (XAI) methods, including SHapley Additive Explanations (SHAP). Unlike traditional statistical approaches, such as linear regression, and opaque AI models like deep neural networks, XAI offers interpretable outputs that help reveal underlying factors contributing to IA. Student demographics, internet use patterns, and psychological variables were analyzed by machine learning algorithms, and SHAP was used to interpret the models. The findings reveal key predictors of IA, such as fear of missing out (FoMO), social anxiety, and social-emotional loneliness. By using XAI, the study shows how specific traits and behaviors elevate IA risk and offers actionable insights for educators and policymakers. It also supports early detection and intervention strategies in educational and psychological contexts.

KEYWORDS
Internet addiction; artificial intelligence; XAI; machine learning

1. Introduction

The rapid expansion of internet access and the proliferation of digital technologies, including social networking sites and smartphones, have led to increased internet usage, particularly among young people. One of the most affected groups is university students, many of whom face challenges related to studying away from their families. While increased access to the internet provides students with valuable educational resources, enhanced communication, and opportunities for academic collaboration and social engagement (Blank & Lutz, 2018), it also raises concerns about potential overuse, manifesting as internet addiction (IA). IA exhibits characteristics commonly associated with substance-related addictions, such as mood regulation, salience, tolerance, withdrawal, conflict, and relapse (Griffiths, 2005; Kurniasanti et al., 2019). In academic literature, IA is frequently used interchangeably with terms like "problematic internet use" and "pathological internet use" and has become a significant concern within the field of behavioral health (Wang & Zeng, 2024). Although widely recognized as a behavioral addiction, IA is not yet formally classified as a diagnostic category in either the DSM-5 or the ICD-11 (Kucukkaraca & Pak, 2023).

Research indicates that young adults, particularly university students, are at elevated risk for IA due to academic pressure, social influences, and developmental or psychological factors that drive extensive online engagement (Gezgin & Akilli, 2016). Studies conducted in Türkiye report the prevalence of problematic internet use among adolescents and young adults to range from 1% to 20% (Alaca, 2020; Kurt & Avci, 2020; Seyrek et al., 2017). In addition, psychological variables such as loneliness (Ayas & Horzum, 2013; Ozdemir et al., 2014; Pontes et al., 2014; Wang & Zeng, 2024), personality traits (Dong et al., 2013; Kayiş et al., 2016; Öztürk et al., 2015), and social interaction anxiety (Çuhadar, 2012; Onyekachi et al., 2022; Weinstein et al., 2015) have been found to be closely associated with IA.

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Despite the extensive research conducted on the psychological underpinnings of IA, including fear of missing out (FoMO), loneliness, social interaction anxiety, and personality traits, the present study adopts a novel methodological approach by integrating Explainable Artificial Intelligence (XAI) into the analysis. Conventional statistical techniques (e.g., regression, structural equation modeling—SEM) often fall short in capturing nonlinear interactions and the individualized effects of variables. In contrast, the use of XAI, particularly SHapley Additive Explanations (SHAP) analysis, enables a more granular, interpretable, and personalized understanding of how psychosocial factors contribute to internet addiction. This approach remains relatively novel in the social sciences, where XAI applications are still emerging but increasingly relevant. By leveraging XAI for model transparency, the study aims to bridge a methodological gap and provide actionable, data-driven insights for early detection and prevention strategies.

Machine learning (ML) techniques offer promising avenues for modeling IA based on psychosocial predictors such as FoMO, loneliness, social interaction anxiety, and personality traits. These models can help identify individuals at risk and inform appropriate support and intervention strategies (Likhith et al., 2024). Recently, the use of ML in the social sciences has gained traction, though existing studies have largely focused on problematic smartphone use (Arpaci, 2023; Elhai et al., 2020b; Raj et al., 2024; Savci et al., 2022). While traditional methods aggregate questionnaire items into composite scale scores and provide reliable and validated measures of psychological constructs, they may not fully reveal the relative predictive importance of each construct in complex behavioral outcomes. ML, especially when combined with explainable AI techniques such as SHAP, complements these psychometric approaches by offering a more detailed understanding of how each scaled variable contributes to IA risk. Its ability to process complex and multidimensional datasets allows for a more nuanced exploration of these relationships.

However, concerns have emerged regarding the use of ML and deep learning for analyzing human behavioral data and supporting professional decision-making, particularly due to their "black box" nature (Kim et al., 2024). To address this, recent studies have emphasized the need for explainable models. XAI, as a response to this need, enhances the interpretability and transparency of AI-based systems (Adadi & Berrada, 2018). In this study, we employ XAI techniques to go beyond mere prediction accuracy and uncover the underlying mechanisms driving IA risk. Although previous studies have investigated the roles of FoMO, loneliness, social interaction anxiety, and personality traits in relation to internet addiction, these constructs have primarily been examined in isolation or through traditional statistical approaches. Given the evolving nature of online engagement among university students, especially in the post-pandemic context, there is a pressing need to revisit these psychological factors using advanced, interpretable methods. By applying explainable AI techniques, this study not only re-evaluates well-established predictors but also uncovers their relative and interactive effects, offering a more contemporary and actionable understanding of internet addiction risk. This approach responds to both methodological and practical gaps in the literature, strengthening the theoretical foundation while enhancing the applicability of the findings in real-world educational and psychological settings. By doing so, we aim to offer insights that are not only statistically robust but also practically useful for educators, counsellors, and policymakers (Stanimirovic et al., 2024).

1.1. Explainable artificial intelligence

XAI refers to a set of processes and methods that allow humans to understand and interpret the decisions made by AI models (Amann et al., 2020). Traditional AI, often considered a "black box" (Preece et al., 2018), provides little insight into how it arrives at a particular outcome, making it difficult to build trust and accountability (Gunning et al., 2019). XAI, however, aims to make AI models more transparent by offering explanations for the model's decisions in human-understandable terms. This interpretability is essential in fields like healthcare, finance, and education, where understanding the "why" behind an AI model's decisions can improve trust, enable error correction, and support regulatory compliance. In short, XAI enhances the usability of AI by ensuring that model predictions are not only accurate but also interpretable. Recently, the use of XAI in studies in the field of social sciences

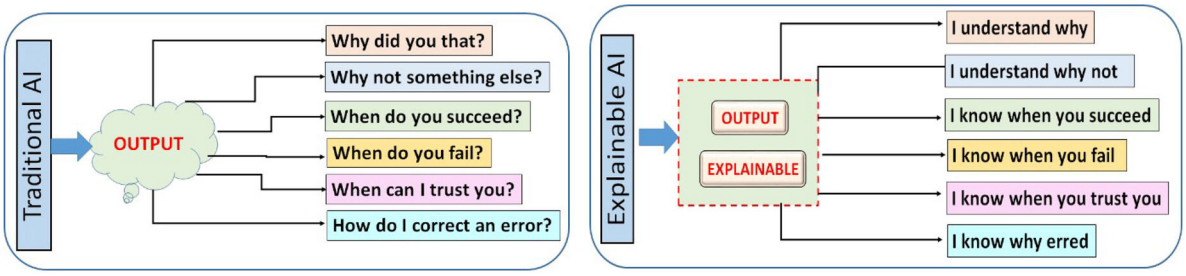

Figure 1. Comparison of traditional AI and explainable AI in terms of model transparency and trust.

has become the subject of attention (Miller, 2019). In Figure 1, a comparison between traditional and XAI is made.

1.2. SHapley additive exPlanations

SHAP is based on cooperative game theory and is one of the popular XAI techniques. Each of the features used for prediction is assigned an importance value. This value indicates how effective the features are in the prediction. This makes it especially valuable in understanding complex machine learning models, as it gives clear, quantifiable insights into how each feature affects the outcome. For example, in a model predicting internet addiction, SHAP can help determine the individual impact of factors like FoMO, loneliness, or social anxiety on the predicted outcome.

1.3. Why do we prefer XAI and SHAP?

Recent concerns have arisen regarding the use of machine and deep learning techniques in fields such as healthcare, engineering, and the social sciences, especially when applied to data relevant to human welfare and therapeutic decision-making (Kim et al., 2024). Despite their high predictive accuracy, these models often lack transparency in how outcomes are generated, leading to challenges in interpretation and trust (Knapič et al., 2021). XAI, through techniques such as SHAP, addresses this issue by enhancing model interpretability. It allows researchers and stakeholders to clearly understand how each input variable contributes to the prediction. This transparency not only fosters greater trust in AI-based decisions but also enables validation and scrutiny of model behavior (Lipton, 2018). Beyond transparency, XAI offers actionable insights for mental health professionals, educators, and policymakers by pinpointing which factors most strongly influence outcomes like internet addiction. Such clarity supports targeted intervention development. Finally, the interpretability of XAI systems facilitates error detection and correction, helping users to identify biases or flawed logic in model outputs and make necessary adjustments.

2. Related work

2.1. Internet addiction and FoMO

FoMO is defined as a contemporary form of addiction that compels individuals to engage extensively with social networks, driven by the anxiety of being uninformed about social developments, particularly following the advent of the internet and smartphones (Gokler et al., 2016; Przybylski et al., 2013). FoMO amplifies the necessity for heightened activity, particularly on social media sites. Individuals experiencing FoMO perpetually engage with the internet and augment their social media usage due to the apprehension of missing online social interactions. This could result in internet addiction over time. Przybylski et al. (2013) asserted that FoMO is a significant predictor of social media engagement and leads individuals to augment their online duration. A study demonstrated a correlation between FoMO and individuals' social attachment requirements and psychological well-being, suggesting that those with a heightened yearning for social connection are more prone to experience FoMO, hence

augmenting their online engagement (Fioravanti et al., 2021). Certain research indicates that FoMO levels are elevated in individuals with internet addiction. Elhai et al. (2020a) asserted that persons with internet addiction allocate more time to social media, hence elevating their levels of FoMO. In these individuals, the compulsion to remain perpetually online induces FoMO, which in turn exacerbates problematic internet technology use. Vaidya et al. (2016) and Anastasya et al. (2022) identified FoMO as a significant risk factor for online addiction, revealing that symptoms of internet addiction were more prevalent among those exhibiting elevated levels of FoMO. These individuals often have adverse effects in their scholastic and social spheres due to their incessant need to remain online. A hypothesis (H_1) was proposed based on literature that elucidates the correlation between internet addiction and FoMO.

H_1: *Individuals with high levels of FoMO tend to develop internet addiction by spending more time on social media platforms.*

2.2. Internet addiction and social and emotional loneliness

Loneliness is commonly defined as an unwelcome and distressing emotional state, often accompanied by feelings such as anxiety, anger, and sadness (Saadati et al., 2021). Although it tends to be more prevalent among adolescents and young adults, loneliness can occur at any stage of life. According to Weiss (1973), loneliness encompasses both emotional isolation and social disconnection. Emotional loneliness refers to the absence of close, meaningful attachments, while social loneliness, often referred to as loneliness in social relationships (LSR), involves lacking a companion with whom to share thoughts, aspirations, and personal experiences. Individuals experiencing loneliness may adopt various coping strategies to address their emotional discomfort, one of which is increased engagement with the internet. Social networking platforms can temporarily alleviate feelings of loneliness by enabling digital social interaction. However, excessive reliance on these platforms may inadvertently contribute to internet addiction (IA), leading individuals to withdraw further from offline relationships (Boursier et al., 2020). Research suggests that individuals facing chronic loneliness are more susceptible to IA, often using online environments as a means to mitigate their sense of isolation (Morahan-Martin & Schumacher, 2000; Odacı & Kalkan, 2010). A review of existing literature consistently shows a significant association between loneliness and IA (Dong et al., 2024; Gu et al., 2023; Karapetsas et al., 2015; Koyuncu et al., 2014; Moody, 2001; Pontes et al., 2014; Saadati et al., 2021). Moreover, several studies highlight that both emotional and social loneliness are particularly predictive of digital addiction, indicating that individuals who feel disconnected in either domain are at greater risk of developing problematic internet use (Altınel et al., 2024; Aslan, 2023; Cui & Yip, 2024; Gezgin et al., 2020; Papapanou et al., 2023; Tatu, 2024; Turan et al., 2023). Drawing upon these findings, the present study formulated Hypothesis 2 (H_2), positing a positive association between loneliness and internet addiction.

H_2: *Individuals with high levels of social and emotional loneliness are more prone to developing internet addiction.*

2.3. Internet addiction and social interaction anxiety

Social interaction anxiety (SIA) refers to a persistent fear or discomfort in social situations, which often leads individuals to avoid face-to-face interactions and seek alternative, less threatening forms of communication (Caplan, 2007). Research has shown that individuals with elevated levels of SIA are particularly vulnerable to IA (Caplan, 2003; Prizant-Passal et al., 2016). For such individuals, online platforms offer a controlled and less judgmental environment for social engagement, reducing the perceived risk of negative evaluation or rejection. The anonymity, flexibility, and emotional distance afforded by digital communication allow socially anxious individuals to bypass the discomfort typically associated with direct interpersonal contact. As a result, they may increasingly turn to online interactions as a coping mechanism, which, over time, can lead to compulsive internet use and eventual addiction (Prizant-Passal et al., 2016). Numerous studies have reported a significant relationship between SIA and

IA (Baloğlu et al., 2018; Hidayat & Nurhayati, 2019; Nwufo et al., 2023; Tosuntaş et al., 2024; Yusuf et al., 2022). In light of these findings, the present study proposed Hypothesis 3 (H_3), which posits a positive association between social interaction anxiety and internet addiction.

H_3: *Elevated social interaction anxiety correlates with an increased propensity for internet addiction.*

2.4. Internet addiction and personality traits

The relationship between IA and personality traits has become a prominent focus in the fields of psychology and behavioral sciences in recent years. Research indicates that individual differences in personality significantly shape internet usage patterns and susceptibility to problematic use, with certain personality profiles exhibiting a higher tendency toward IA across domains such as academic activities, social networking, and online gaming (Miskulin et al., 2022). Within the framework of the Big Five personality model such as Neuroticism (NEU), Extraversion (EXT), Openness to Experience (OPN), Agreeableness (AGR), and Conscientiousness (CONS) (Goldberg, 1990; McCrae & Costa, 1987), evidence suggests that individuals high in NEU are more likely to use the internet as a coping mechanism for stress and negative emotions, making them more vulnerable to addiction (Di et al., 2019; Kuss et al., 2013; Mak et al., 2021; Stodt et al., 2018). In contrast, individuals high in CONS and AGR tend to exhibit more self-regulated internet behavior and show a lower risk of IA (Gunaydin, 2021; Jojo & Sundaramoorthy, 2022; Ksiksou et al., 2023).

The role of EXT in predicting IA is more nuanced. While extraverted individuals often prefer face-to-face interaction and are generally less prone to IA, their impulsivity and attraction to novelty may increase their susceptibility to specific online activities (Kayiş et al. 2016; Kuang et al., 2020; Xiao et al., 2019). Similarly, the influence of OPN appears to be dual in nature. Some studies suggest that individuals with high openness, characterized by imagination, creativity, and intellectual curiosity, engage in a wide range of offline activities that reduce excessive internet use. Others argue that their affinity for novel stimuli may drive increased online engagement, particularly in exploratory or creative digital environments (Celik et al., 2012; Miskulin et al., 2022; Öztürk et al., 2015; Servidio, 2014). These findings underscore the importance of considering personality profiles when examining patterns of internet use and developing tailored interventions. Accordingly, Hypothesis 4 (H_4) was formulated based on the existing literature to investigate the predictive role of personality traits in IA.

H_4: *Personality traits influence the likelihood of developing internet addiction.*

3. Methodology

3.1. Participants and procedure

In April 2024, we administered an online survey at a state university located in a border city in Türkiye. Institutional Review Board (IRB) approval was obtained prior to data collection, and all procedures adhered to the principles of the Declaration of Helsinki. Participants were recruited through the voluntary involvement of students enrolled in the "*Addiction and Struggle Against Addiction-I*" course offered at the university. This elective course, open to all departments, focuses on the psychological, social, and biological dimensions of addiction as well as prevention strategies. Participation in the study was entirely voluntary. Students were informed that their responses would be anonymized and used solely for research purposes. Importantly, no course credit, bonus points, or other incentives were offered for survey completion. The researchers developed a questionnaire and collected responses using Google Forms. All measures were administered in Turkish. A total of 1,180 undergraduate students participated in the study. The mean age of the participants was 21.75 years (age range = 19–34, SD = 2.62). Of these, 522 (44.2%) identified as female and 658 (55.8%) as male. The participants were enrolled in 35 different undergraduate departments. The participants were primarily enrolled in mechanical engineering ($n = 252$, 21.4%), medicine ($n = 181$, 15.3%), and computer engineering ($n = 173$, 14.7%). The remaining students ($n = 574$, 48.6%) represented a broad range of disciplines, including

health management, physical therapy and rehabilitation, physics, chemistry, mathematics, nursing, electrical and electronic engineering, dentistry, Turkish language and literature, architecture, food engineering, genetics, econometrics, economics, and business administration. Further demographic and departmental details are presented in Table 1.

3.2. Instruments

3.2.1. Young's internet addiction scale-short form (YIAS-SF)

The Young's internet addiction scale-short form (YIAS-SF), developed by Young (1998) and abbreviated by Pawlikowski et al. (2013), is a five-point Likert-type scale with 12 items. Kutlu et al. (2016) performed the Turkish adaption of the FCAS on adolescents and university students. Exploratory component analysis (EFA) revealed that the scale included a singular component in both university students and adolescents. For college students, the scale had a Cronbach alpha reliability score of 0.91, and for teens, it was 0.86. It was found that the scale had a test–retest reliability score of 0.93 for college students and 0.86 for teens. There are no things on the test that are scored backwards and high scores mean that you are more likely to become addicted to the internet. For the university students in the study, the Cronbach alpha dependability score was found to be.90.

3.2.2. Fear of missing out scale

The FoMOs was created by Przybylski and colleagues in 2013. The assessment has 10 items addressing anxiety related to the absence from social occasions, with response possibilities ranging from "1 = Not at all true of me" to "5 = Extremely true of me." Elevated scores indicate a greater likelihood of persons experiencing fear of missing out. "I fear that others have more fulfilling experiences than I do" is a prime example of the FoMO Scale. We utilized the previously validated Turkish version (Gokler et al., 2016). The reliability coefficient of the original scale, determined using Cronbach's alpha, is 0.95. The internal consistency of our entire sample was 0.86.

3.2.3. The Big Five Inventory

The 10 items personality scale created by Rammstedt and John (2007), assessing five fundamental personality qualities, was adapted to Turkish culture and language by Horzum et al. (2017). The five primary personality qualities are extraversion, agreeableness, conscientiousness, neuroticism, and openness to experience. With values between.81 and.90, the sub-dimensions of the Big Five Inventory (TBFI) in this study were found to have good internal consistency dependability. The Cronbach's alpha values for the scale's factors varied from 0.71 to 0.77 in the study.

3.2.4. Social interaction anxiety scale)

This instrument, created by Leary and Kowalski (1993) and modified into Turkish by Coskun (2009), is a unidimensional scale consisting of 15 elements. The SIAs assesses social interaction anxiety using a five-point Likert scale, with answer possibilities from 1 ("strongly disagree") to 5 ("strongly agree"). Five items on the scale are assessed negatively and thus reverse scored. Elevated scores on the SIAS

Table 1. Participant characteristics (N = 1180).

Categories	Frequency	Mean (SD)	%
Gender			
Male	658		55.8
Female	522		44.2
Age (range 18–48 years)		21.75(2.62)	
Department			
Mechanical engineering	252		21.4
Medicine	181		15.7
Computer engineering	173		15.3
Health management	42		3.6
Econometric	41		3.5
Physiotherapy and rehabilitation	38		3.2
Others	453		37.3

Note: In Table 1, the departments with 30 or less participants under the heading "Department" are grouped under "others."

signify heightened social interaction anxiety. All items in the scale have factor loadings over 0.45. The overall Cronbach's alpha coefficient of the scale is 0.91, signifying acceptable reliability (Coskun, 2009). The dependability coefficient, measured by Cronbach's alpha, was determined to be 0.72 in the study.

3.2.5. Social and Emotional Loneliness Scale for Adults

The Social and Emotional Loneliness Scale for Adults (SELSA-S), created by DiTommaso et al. (2004) and modified into Turkish by Çeçen (2007), was employed to assess the social and emotional loneliness of university students. The scale has 15 items and employs a seven-point Likert-type format, ranging from "not entirely suitable for me" to "completely suitable for me." The scale comprises three sub-dimensions: loneliness in social, romantic, and family connections. Cronbach's alpha values between 0.74 and 0.88 were recorded in the reliability and validity tests for the measure. In this study, the Cronbach's alpha coefficients for the sub-dimensions of SELSA-S were 0.80, 0.82, and 0.79.

3.3. Data collection and data analysis process

The data collection process for the study took place over 2 months, during December and January of the 2023–2024 academic year. Data were collected through Google Forms based on voluntary participation, and ethical approval was obtained for the study. At the beginning of the survey, students were required to complete a consent form confirming their voluntary participation. For data analysis, traditional methods such as descriptive statistics were used, and Pearson correlation coefficients were calculated to examine the relationships between variables. However, as the study primarily aimed to predict the prevalence of IA among university students, XAI methods were employed for prediction purposes.

3.4. Model constructions

In the study, firstly, there is a model building phase. In this context, the model was created using five popular (new/recently used/current) machine learning methods. These methods are gradient boosting, XGBoost, random forest, SVR, and LightGBM algorithms, respectively. The working principles of the algorithms are given below.

3.4.1. Machine learning algorithm

3.4.1.1. Gradient boosting. This algorithm iteratively combines multiple weak learners, usually decision trees, to create a strong model. The core idea behind gradient boosting is to minimize the residual error of the previous model by fitting a new model on the errors. This process is done in a stage-wise manner, where each new model corrects the mistakes made by the ensemble of previous models. Gradient boosting optimizes the loss function using gradient descent, adjusting the model parameters to reduce the overall prediction error. The gradient boosting method provides the ability to handle various types of data (e.g., continuous, categorical). It also provides the flexibility in optimizing different loss functions for both regression and classification tasks (Friedman, 2001).

3.4.1.2. Extreme gradient boosting. This algorithm is specifically designed to improve prediction accuracy and efficiency in structured/tabular data problems. It leverages an ensemble technique where multiple weak learners, usually decision trees, are combined to create a strong predictive model. The key innovation of XGBoost lies in its ability to optimize both the model's predictive performance and its computational efficiency through advanced regularization techniques (L1 and L2 regularization), tree pruning, and parallelization. The algorithm minimizes a loss function using gradient descent, iteratively refining the model by correcting errors made by the previous trees. XGBoost has gained prominence due to its robustness in handling missing values, its ability to scale to large datasets, and its superior performance in competitive machine learning scenarios (Chen & Guestrin, 2016).

3.4.1.3. Random forest. After creating a collection of decision trees, the algorithm combines the predictions and tries to improve the model performance. The training set is divided into random subsets and each of the trees in the created forest is trained on these subsets. By using techniques like bootstrapping

(sampling with replacement) and random feature selection at each node, random forest mitigates the risk of high variance that is often seen in individual decision trees. The final prediction is obtained by averaging. This diversity of trees leads to a more accurate model that can handle complex datasets with minimal tuning (Breiman, 2001).

3.4.1.4. Support vector regression. Support vector regression (SVR) is a machine learning algorithm that aims to find the function that best fits the given data while maintaining generalization. The primary goal of SVR is to find a function that deviates from the actual data points by no more than a specified margin (epsilon), while also minimizing the model's complexity. SVR achieves this through the use of kernel functions, which map input features into higher-dimensional spaces, enabling it to model nonlinear relationships. This algorithm is offering a robust solution even in the presence of noisy data (Vapnik, 1995).

3.4.1.5. LightGBM. It is an efficient and scalable algorithm. The goal is to correct the mistakes of the previous tree. This process continues iteratively. Unlike traditional gradient boosting methods that grow trees level by level, light gradient boosting machine (LightGBM) constructs trees leaf-wise, meaning it grows the leaves that minimize the error the most, resulting in a deeper, more complex tree. This approach reduces overfitting and increases the model's predictive power, especially in large datasets. Additionally, LightGBM employs techniques such as histogram-based decision making and data parallelism, which further enhance its speed and performance, making it a powerful tool for regression tasks. By using a histogram approach, it groups continuous features into discrete bins, significantly reducing memory usage and speeding up the training process (Ke et al., 2017). This makes LightGBM a popular algorithm for real-world regression problems.

3.4.2. Hyperparameter optimization: GridSearchCV

One of the factors that affect the success of the algorithms is the ability to make accurate adjustments to the model's hyperparameters. Grid Search is an effective optimization method used in this hyperparameter tuning process. Grid Search tries to find the best combination by performing a systematic search among the possible combinations of values of the specified hyperparameters. In this respect, the hyperparameters were found using Python's GridSearchCV function. Table 2 shows the tuning parameters considered in GridSearchCV.

3.5. Model evaluation

At this stage, the dataset is usually divided into training and test sets in a certain proportion. After the training process, the model is tested on the test set. However, when the model is tested with this method, it is tested on a specific dataset. Moreover, the success of the model varies according to the separation rate. A frequently encountered problem in machine learning methods is the over fitting problem. In the over-fitting problem, the algorithm performs quite well in training but performs poorly

Table 2. Tuning parameters considered in GridSearchCV.

Algorithm	n_estimators	Learning_rate	Max_depth	Best parameters
GBM	50, 100, 150	0.05, 0.1, 0.2	3,4,5	'n_estimators': 50, 'learning_rate': 0.05, 'max_depth': 3,
XGBoost	50, 100, 150	0.05, 0.1, 0.2	3, 4, 5	'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 50
Random Forest	100, 150, 200, 250	'max_features': ['auto', 'sqrt', 'log2'],	None, 5, 10, 20	'max_depth': 5, 'max_features': 'auto', 'n_estimators': 250
SVR	'C': [0.1, 1, 10]	'epsilon': [0.1, 0.2, 0.3]	'gamma': ['scale', 'auto'],	'C': 1, 'epsilon': 0.3, 'gamma': 'auto'
LightGBM	100, 150, 200	0.05, 0.1, 0.2	3, 4, 5	'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 100

in testing. In other words, the algorithm may make incorrect predictions when faced with an unknown dataset. The best way to detect the overfitting problem is to test the model on more data. In this case, the cross-validation model evaluates on more than one validation set. Thus, cross-validation provides a more realistic estimate of the generalization performance of the model. For this reason, 10-fold cross-validation method was preferred in the data processing stage of the study. In the cross-validation method, the dataset is divided into k subsets (layers/fold). Subsequently, training is performed on all subsets. However, one subset (k-1) is left for the evaluation of the trained model. In this method, k iterations are performed, each time leaving a different subset for testing. The number k is usually 10, as in this study.

3.5.1. Model performance evaluation

The performance metrics used in the performance analysis of the model are listed below.

R-Squared (R^2): R^2 is represents the proportion of the variance in the dependent variable that is predictable from the independent variables. It compares the sum of squared residuals (errors between the observed and predicted values) to the total sum of squares (the variance of the observed data). If the R^2 value is close to 1, the algorithm is considered to explain most of the variance in the data, otherwise it is considered to make poor predictions.

It is calculated using Eq. (1).

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(z_i - \hat{z}_i)^2}{\sum_{i=1}^{n}(z_i - \overline{z})^2} \tag{1}$$

Where: z_i is the actual value, $\hat{z}_i$ is the predicted value, $\overline{z}$ is the mean of the actual values, and n is the number of data points.

Root-mean-squared error (RMSE): This metric is commonly used in regression algorithms, calculated by taking the square root of the average of the squared differences between the observed true results and the predictions. A lower RMSE indicates a better fit of the model to the data. It is calculated using Eq. (2).

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(z_i - \hat{z}_i)^2} \tag{2}$$

Mean absolute error (MAE): MAE represents the average absolute differences between the observed actual outcomes and the predictions. Unlike RMSE, MAE does not square the errors, so it gives equal weight to all errors, regardless of their magnitude. This makes MAE a simpler and more intuitive measure of model accuracy, especially when large errors are not of particular concern (Hastie et al., 2009). A smaller MAE indicates better model performance, and because it is linear (not squared), it is less sensitive to outliers compared to RMSE. It is calculated using Eq. (3).

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|z_i - \hat{z}_i| \tag{3}$$

In the study, the performance metrics of the model for training and 10-fold cross-validation were calculated separately. Subsequently, the model evaluation was performed.

3.6. Model interpretation with the SHAP method

In this study, SHAP analysis was used to understand complex machine learning models. Thus, SHAP analysis provides both local and global interpretabilities. Depence plot, Importance plot, and Beeswarm plot were used for global interpretability. For local interpretability, Waterfall plot and Force plot were used. The procedure of the study is shown in Figure 2.

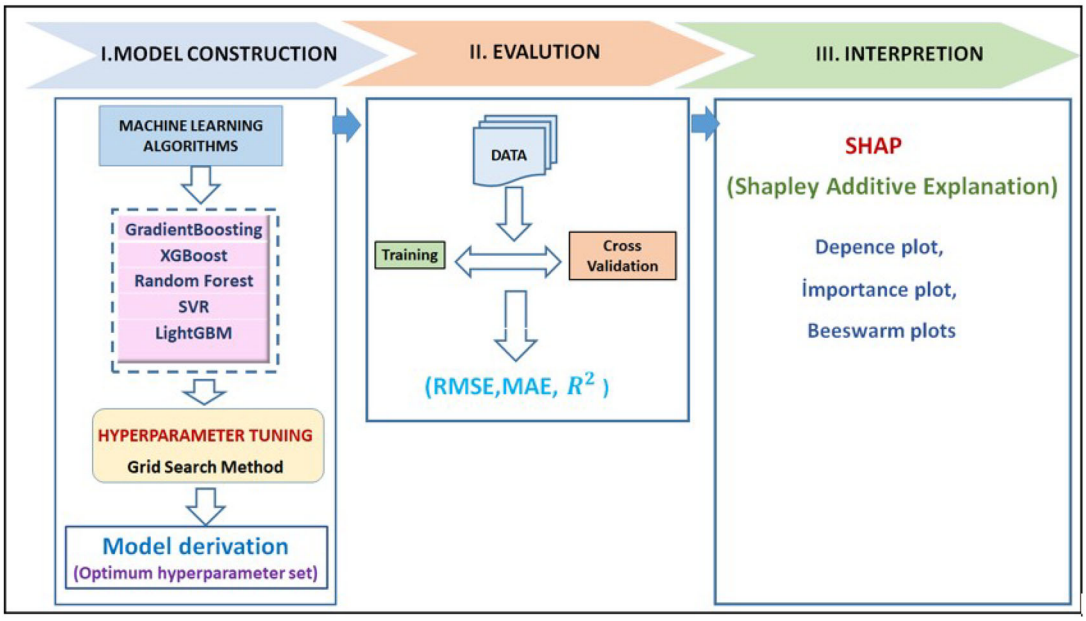

Figure 2. Flow diagram of the study.

Table 3. Zero-order bivariate correlations between IA, FoMO, SIA, multidimensional loneliness, personality traits.

Variables	1	2	3	4	5	6	7	8	9	10	11
1.IA	1										
2.FoMO	0.597**	1									
3. SIA	0.449**	0.448**	1								
4.LSR	0.269**	0.128**	0.192**	1							
5.LRR	0.113**	0.131**	0.139**	0.169**	1						
6. LFR	0.316**	0.212**	0.195**	0.555**	0.136**	1					
7.EXT	−0.236**	−0.143**	−0.436**	−0.391**	−0.265**	−0.268**	1				
8.AGR	−0.311**	−0.258**	−0.204**	−0.259**	−0.041	−0.288**	0.137**	1			
9.CONS	−0.317**	−0.202**	−0.274**	−0.257**	−0.143**	−0.282**	0.388**	0.208**	1		
10.NEU	0.166**	0.148**	0.214**	0.140**	0.047	0.123**	−0.187**	−0.074*	−0.116**	1	
11.OPN	−0.141**	−0.066*	−0.107**	−0.149**	−0.058*	−0.150**	0.186**	0.139**	0.244**	−0.060*	1

Note: $^{**}p < 0.01$; $^{**}p < 0.05$. IA: Internet addiction; FoMO: Fear of Missing Out; SIA: Social Interaction Anxiety; LSR: Loneliness in Social Relationships; LRR: Lonelines in Romantic Relationships; LFR: Loneliness in Family Relationships; EXT: Extraversion; AGR: Agreeableness; CONS: Conscientiousness; NEU: Neuroticism; OPN: Openness to experience.

4. Results

4.1. Correlational analysis results of internet addiction, fear of missing out, loneliness, social interaction anxiety, and personality traits

Table 3 presents the correlations between IA, FoMO, SIA, different dimensions of loneliness (Loneliness in Social Relationships—LSR, Loneliness in Romantic Relationships—LRR, Loneliness in Family Relationships—LFR), and personality traits (Extraversion—EXT, Agreeableness—AGR, Conscientiousness—CONS, Neuroticism—NEU, and Openness to Experience—OPN). IA is strongly and positively correlated with FoMO ($r = 0.597$, $p < 0.01$), indicating that individuals with higher FoMO levels tend to have higher levels of IA. Similarly, SIA is significantly and positively associated with IA ($r = 0.449$, $p < 0.01$), suggesting that individuals experiencing higher levels of SIA are at a greater risk of IA. Regarding the subdimensions of loneliness, IA has the strongest correlation with loneliness in family relationships ($r = 0.316$, $p < 0.01$), followed by loneliness in social relationships ($r = 0.269$, $p < 0.01$) and loneliness in romantic relationships ($r = 0.113$, $p < 0.01$). This finding suggests that individuals who feel lonely within their family environment may be more prone to IA compared to those who experience loneliness in social or romantic contexts. In terms of personality traits, extraversion is negatively correlated with IA ($r = -0.236$, $p < 0.01$), indicating that more extraverted individuals tend to have lower levels of IA. Similarly, agreeableness ($r = -0.311$, $p < 0.01$) and conscientiousness ($r = -0.317$, $p < 0.01$) are also negatively associated with IA, suggesting that individuals who are more

Table 4. Performance metrics of the algorithms in training and 10-fold cross-validation.

	Training			10-fold cross validation		
Algorithm	R^2	RMSE	MAE	R^2	RMSE	MAE
Gradient Boosting	0.53	6.45	5.02	0.43	7.09	5.33
XGBoost	0.53	6.46	5.02	0.43	7.12	5.36
Random Forest	0.59	5.98	4.63	0.45	7.00	5.17
SVR	0.45	6.95	5.17	0.44	7.03	5.25
LightGBM	0.56	6.24	4.79	0.42	7.15	5.30

agreeable and conscientious are less likely to develop IA. On the other hand, neuroticism exhibits a positive correlation with IA ($r = 0.166$, $p < 0.01$), meaning that individuals with higher emotional instability are at greater risk of IA. Lastly, openness to experience is negatively related to IA ($r = -0.141$, $p < 0.01$), implying that individuals with higher openness levels may have lower tendencies toward IA. The findings indicate that IA is significantly associated with FoMO, SIA, and multidimensional loneliness, as well as personality traits such as extraversion, agreeableness, and neuroticism.

4.2. Model evaluation results

The R^2, RMSE, and MAE values obtained in training and 10-fold cross-validation are given in Table 4.

When examining Table 4, it is observed that the highest R^2 value during training is 0.59, obtained using the random forest algorithm. Conversely, the lowest R^2 value is 0.45, derived from the SVR algorithm. In 10-fold cross-validation, the highest R^2 value was also obtained using the random forest algorithm, with a value of 0.45. When evaluating the training and 10-fold cross-validation performances separately, the R^2 value of 0.59 obtained from the random forest algorithm during training indicates that the model explains 59% of the variance in the training data. This suggests that the model has effectively learned complex relationships and fits the training data well. On the other hand, the R^2 value of 0.45 obtained from the SVR algorithm suggests a comparatively lower performance, indicating that the model explains only 45% of the variance in the training data, demonstrating weaker model fit compared to other algorithms. The R^2 value of 0.45 obtained in 10-fold cross-validation serves as a crucial indicator of the model's generalizability. While the random forest algorithm exhibits high performance on training data, the lower cross-validation results suggest that its performance decreases when applied to new, unseen data. Additionally, the lower performance of the SVR algorithm during training implies that its capacity to learn relationships within the data is limited, which may result in even lower performance in cross-validation.

4.3. Model interpretation

Since the random forest algorithm showed the best performance among the algorithms used in the study, global interpretation was used to understand the model created with this algorithm and to determine the most important features that guide its predictions.

4.3.1. Global feature interpretation

Global interpretation methods are used to describe the expected behavior of a machine learning model over the entire distribution of input variables. The goal of XAI at the global level is to describe the behavior of the model across the entire dataset. This provides insight into the main factors influencing the model and the general trends and patterns observed. Take, for example, the case of identifying the key factors contributing to technology addiction among university students. Global interpretations will highlight the primary determinants influencing addiction risk across the entire sample and provide insights into potential intervention strategies. In this context, the study will reveal the variables that predict IA of university students and their predictive value, and as a result, it will be interpreted what can be done through these variables in the struggle against IA.

4.3.1.1. Analysis of the importance and contributions of model features. In the SHAP analysis used for global interpretation, the mean absolute SHAP value for each feature across the entire dataset is

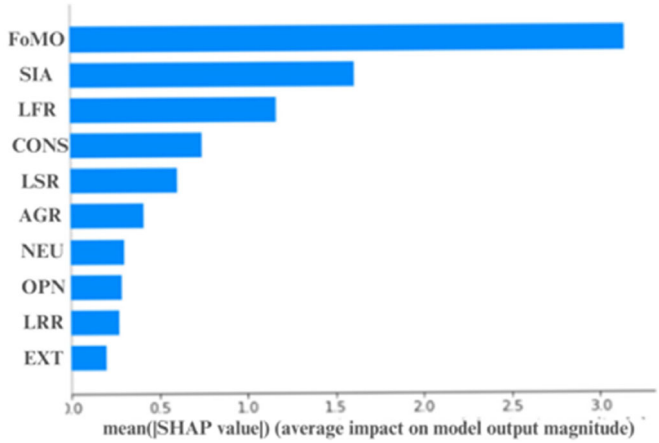

Figure 3. SHAP summary plots for internet addiction predictors.

examined. This value represents the magnitude of each feature's contribution to the predicted IA value, either in a positive or in a negative direction. Features with higher mean absolute SHAP values are considered more predictive. Compared to traditional feature importance metrics, the use of these values in feature importance graphs offers several advantages:

1. *Theoretical rigor:* Mean absolute SHAP values provide a more rigorous measure of feature importance than traditional methods.
2. *Model-agnostic interpretation:* While traditional feature importance measures indicate how much each feature improves the model's predictive performance in an algorithm-specific manner, SHAP values focus on identifying which features have the most influence on predictions.

As illustrated in Figure 3, mean absolute SHAP values are typically visualized as bar charts that rank features by their importance. The x-axis of the graph represents the contribution of each feature to the prediction outcome, calculated as the average of the absolute SHAP values. According to this visualization, the feature with the highest contribution is FoMO, followed by SIA, while EXT is identified as the feature with the lowest contribution.

4.3.1.2. Beeswarm graphics. In this study, another plot, the Beeswarm graph, was used to understand the importance or contribution of the features for the whole dataset. Indeed, Beeswarm graphs can be used to reveal the relative importance of the features as well as their actual relationship with the predicted result. When Figure 4 is examined, each point in the graph shows the Shapley value calculated for each row of data for the relevant attribute. For each variable, each sample of the dataset appears as

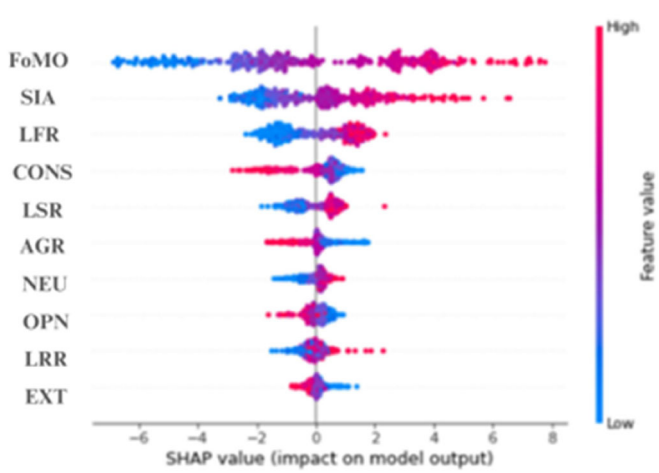

Figure 4. Beeswarm graphics.

its own point in the beeswarm graph. According to this graph, the points are expected to be distributed horizontally, if the points are stacked vertically, the SHAP value is high density here. As a result, each cell used in model training yields as many Shapley values as the number of data points. Looking again at Figure 4, the *y*-axis indicates the attributes used in model training. As can be seen, the attribute that has the most impact on the result is ranked from top to bottom in order of significance. As a result, according to the graph, FoMO is the most important variable that determines IA. The least impact on IA is seen to be the EXT personality trait variable.

In addition, positive values on the X-axis are those that contribute positively to the IA forecast. Negative values are the values that contribute negatively to the prediction. The colors of the points on the X-axis allow us to comment on the numerical value of the data. It can be said that the numerical value of the data increases as it approaches from blue to red. In the FoMO row, an increase in the red dots indicates the possibility of an increased risk of internet addiction. Also, in Figure 4, we can see, for example, that higher values of FoMO have positive SHAP values (dots moving to the right gradually turn red) and lower values of FoMO have negative SHAP values (dots moving to the left gradually turn blue). This suggests that individuals suffering from FoMO are more likely to have IA. This is the similar for SIA and LFR variables, respectively.

Beeswarm graphs are information dense and provide an overview of SHAP values for many features at the same time. However, to really understand the relationship between the values of a feature and the predicted results of the model, it is necessary to examine the dependency plots.

4.3.1.3. Feature dependence graphics. In the study, the SHAP dependency graph was conducted to observe the interactions between input variables in terms of their contributions. The most evident trend among the results of the SHAP dependency graph is associated with the FoMO and SIA variables (graphs), which are ranked high in the contribution rankings based on the SHAP pivot graph. Graph values and variables are shown in Figure 5.

The dependency graph shows the effect of a single attribute on the model. The values on the y-axis on the left show the Shapley values that positively and negatively affect the result. The data's real values are shown on the x-axis. The attribute value (SIA) on the right *y*-axis represents the feature that interacts most with the attribute we interpreted (FoMO). Red colored dots indicate high interaction and blue-colored dots indicate low interaction. According to the graph, it can be said that as the SIA value increases, the probability of FoMO increases.

5. Discussion

In this study, XAI methods were used to predict university students' tendencies towards IA. The study used various machine learning algorithms to determine the impact of psychosocial factors such as

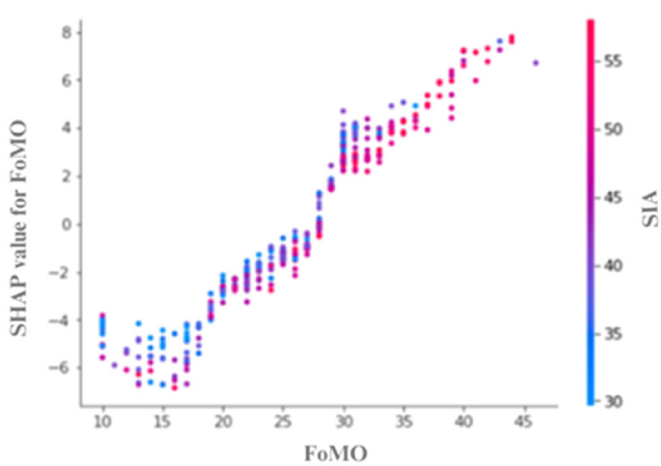

Figure 5. Feature dependence graph.

FoMO, SIA, multidimensional loneliness and personal traits on IA, and interpreted the results with SHAP method.

The findings indicate that FoMO is the most significant predictor of IA among university students. SHAP analysis demonstrates that higher FoMO levels considerably increase the predicted risk of IA (see Figures 3–5). Thus, Hypothesis 1 was supported. This finding aligns with previous research suggesting that FoMO encourages individuals to stay constantly online, leading to excessive social networking sites and internet use (Anastasya et al., 2022; Elhai et al., 2020a; Giancola et al., 2024; Kargın et al., 2020; Setyaningsih et al., 2023; Vaidya et al., 2016). Being members of Generation Z, university students may be particularly vulnerable to digital stressors. University is a critical transition period from adolescence to adulthood, during which individuals assume both academic and life responsibilities while being away from their families. Students living apart from their families may experience increased psychological fragility during this time. Furthermore, considering that the prefrontal cortex, responsible for decision-making and self-regulation, is still underdeveloped during this period, experiencing negative emotional states is highly probable. Social networking sites and smartphones often serve as coping tools for young individuals. FoMO may function as an escape mechanism, exacerbating excessive internet and social media use by exploiting emotional vulnerabilities (Meynadier et al., 2025). Previous studies indicate that elevated levels of FoMO correlate with the heightened psychological discomfort and, more specifically, diminished psychological well-being (Metin-Orta, 2020; Wu et al., 2013). Research indicates that those who often compare themselves to others on social media and endeavour to uphold a favourable online persona are at an elevated risk of depression (Robinson et al., 2019). Given that online social comparison is a defining feature of individuals with high FoMO, it can be asserted that FoMO actively influences the association between IA and psychological distress. Therefore, interventions targeting IA should incorporate strategies to enhance psychological well-being in order to mitigate the impact of FoMO and curtail excessive internet usage. Although the current findings suggest a strong link between FoMO and increased social media use, leading to higher IA risk, it is important to acknowledge that not all forms of social media engagement are inherently problematic. Recent studies have shown that platforms such as TikTok are being explored for active and engaging educational practices (Wang et al., 2024). In this context, XAI methods such as SHAP may offer additional value by helping to differentiate between beneficial and maladaptive usage patterns. Future studies could extend this framework to distinguish specific social media behaviors that foster academic growth versus those contributing to compulsive usage, thus offering more nuanced intervention strategies.

This study identified a positive relationship between university students' levels of social and emotional loneliness and IA. The SHAP analysis (Figure 3) reinforces this finding, as all dimensions of loneliness (LFR, LSR, LRR) showed positive SHAP values, indicating that increased loneliness is associated with a higher risk of IA. Machine learning algorithms revealed that among the three subdimensions, and LFR was the strongest predictor of IA, followed by LSR and LRR. Thus, Hypothesis 2 is supported. These findings are consistent with previous literature. For instance, Iskender (2018) demonstrated that social loneliness and familial emotional loneliness are significant predictors of IA. Similarly, Tabak and Zawadzka (2017) found a significant relationship between IA and emotional loneliness among Polish students. Sart and Sezgin (2022) also suggested that weak family ties increase loneliness, prompting individuals to seek emotional support online, potentially contributing to IA. In the context of problematic social media use, a fundamental component of internet addiction, previous studies have shown that students who live with their families exhibit the lowest levels of social media addiction (Uyaroğlu et al., 2022; Ozdemir, 2019). Gezgin and Ümmet (2021) suggested that increased feelings of loneliness within family relationships raise the risk of nomophobia, which is defined as the fear of being separated from one's smartphone. This finding suggests that young individuals living with their families receive greater social support, which may reduce their reliance on social networking sites compared to those living alone (Uyaroğlu et al., 2022). In conclusion, the findings of this study align with previous research highlighting the impact of loneliness on internet addiction and underscore the crucial role of family support in mitigating digital addiction.

These findings also suggest that universities could play a proactive role in mitigating familial loneliness among students, especially those living away from home. Beyond traditional counselling services,

the integration of AI-driven support systems, such as chatbots or predictive monitoring tools, could provide scalable and responsive interventions. However, the success of such implementations depends on students' acceptance and engagement with these systems. Recent research highlights that institutional support, emotional engagement, and playfulness significantly influence students' acceptance of AI-based technologies in higher education (Wang et al., 2025). In this regard, universities should ensure that support mechanisms addressing loneliness are not only accessible but also perceived as trustworthy, empathetic, and aligned with students' digital habits. Embedding human-centered design principles into AI-driven tools may help overcome emotional barriers and increase utilization among students at risk of internet addiction due to emotional disconnection from family environments.

Another key finding of this study is that, after FoMO and social–emotional loneliness, SIA demonstrates the second-highest explanatory power in predicting IA, indicating a positive relationship. The SHAP analysis (Figure 3) further supports this finding, as SIA emerged as one of the strongest predictors of IA, with high SHAP values consistently pushing IA scores upward. This indicates that individuals with higher levels of SIA levels are significantly more likely to develop IA. Accordingly, the Hypothesis 3 was supported. These findings align with previous studies in the literature (Çuhadar, 2012; Onyekachi et al., 2022; Traş & Gökçen, 2020; Weinstein et al., 2015; Yücens & Üzer, 2018). Moreover, this result corroborates earlier research suggesting that individuals with high levels of social anxiety tend to avoid offline interactions and instead use the internet as a coping mechanism (Caplan, 2007; Prizant-Passal et al., 2016). The SHAP summary and beeswarm plots illustrate that as SIA values increase (moving from blue to red), IA risk rises, reinforcing the role of online engagement as a coping strategy for socially anxious individuals. Similarly, prior studies indicate that individuals with internet addiction often rely on online environments to manage underlying psychological and developmental difficulties (Ko et al., 2006). Socially anxious and lonely individuals often seek social support *via* social networking sites to compensate for a lack of face-to-face interaction (O'Day & Heimberg, 2021).

This is also reflected in the SHAP dependence plot, where SIA shows a strong interaction effect with FoMO, indicating that socially anxious individuals may experience heightened FoMO, further driving their excessive internet use. Taken together, these findings suggest that individuals experiencing high levels of SIA are more prone to excessive social networking site use and internet, thereby increasing their vulnerability to IA. This insight can inform tailored intervention strategies. For instance, mental health professionals may prioritize combined cognitive–behavioral interventions that target both FoMO-driven compulsions and social anxiety symptoms. Educators could implement peer-led support programs or structured group activities that reduce social anxiety in safe offline environments, while also fostering awareness around the emotional triggers of FoMO. This dual-target approach may be more effective than addressing these risk factors in isolation. This actionable insight, derived from the SHAP dependence plot, provides added value beyond traditional regression models, which generally estimate the main effects but often overlook nonlinear interactions between variables. While a regression model may identify both FoMO and SIA as independent predictors of internet addiction, it would likely fail to reveal that their combined effect is disproportionately higher than the sum of their individual effects. In contrast, SHAP's local and global interpretability reveals that students with co-elevated FoMO and SIA scores constitute a particularly high-risk subgroup. This enables educators and mental health professionals to move from generalized interventions toward precisely targeted strategies. For instance, universities might develop dual-modality programs that simultaneously address social anxiety through exposure-based interventions while mitigating FoMO through psychoeducation and digital literacy training (Gurkan et al., 2022). Furthermore, policy designers could use these insights to tailor preventive efforts for subpopulations with overlapping psychological vulnerabilities, thereby improving the allocation of limited mental health resources.

While XAI contributes to greater transparency and interpretability in predictive models, its application in sensitive domains such as psychological profiling of students must be approached with ethical caution. Predicting internet addiction using AI raises concerns about data privacy, the risk of stigmatizing students identified as "at risk," and the unintended consequences of automated labeling. These issues are particularly salient in educational contexts, where students may be vulnerable to misinterpretation or institutional misuse of predictive outputs. To align with ethical best practices, future

implementations of XAI-driven interventions should prioritize informed consent, ensure anonymity, and involve mental health professionals in interpreting risk scores. Moreover, predictive outputs should not be used for punitive purposes but rather to guide supportive and voluntary interventions grounded in empathy and respect for student autonomy.

Finally, the analysis revealed significant relationships between all five personality traits and IA (Celik et al., 2012; Kuss et al., 2014; Landers & Lounsbury, 2006; Samarein et al., 2013), thereby supporting the Hypothesis 4. SHAP analysis showed that CONS and AGR had the highest absolute SHAP values, making them the strongest predictors of IA, indicating that individuals with higher levels of CONS and AGR are less likely to develop IA. This finding is in line with previous research showing that individuals with high levels of CONS are more capable of regulating internet use, maintaining self-discipline, and prioritizing offline responsibilities over online engagement (Gunaydin, 2021; Lee & Lim, 2021). Negative SHAP scores for CONS further highlight the role of goal-directed behavior and self-control in mitigating problematic internet use. Likewise, AGR demonstrated a strong protective influence. Individuals with high AGR scores may benefit from stronger social support systems and prefer face-to-face interactions, resulting in healthier digital habits (Miskulin et al., 2022). In contrast, individuals lower in AGR tend to favor online over offline interactions (Kuss et al., 2013), potentially increasing their vulnerability to IA.

NEU, while still an important predictor, ranked below CONS and AGR in absolute SHAP importance. However, in contrast to CONS and AGR, NEU had positive SHAP values, suggesting that individuals with higher NEU scores were more prone to IA. This is consistent with previous literature suggesting that emotionally unstable individuals may resort to excessive internet use as a coping mechanism for stress, anxiety, and negative emotions (Di et al., 2019; Kuss et al., 2014; Miskulin et al., 2022). The SHAP analysis also reveals that EXT has the lowest effect on IA, suggesting that EXT may not play an important role in predicting IA. While some studies suggest that extraverts are less prone to IA due to their preference for face-to-face interactions, our findings suggest that this trait has only a small protective effect (Kayiş et al., 2016; Kuang et al., 2020). Similarly, openness to OPN exhibited weak SHAP contributions, suggesting that intellectual curiosity and novelty-seeking tendencies do not significantly influence IA risk. This is consistent with studies showing that individuals with high levels of OPN may enjoy exploring new online content, but this does not necessarily lead to problematic use (Öztürk et al., 2015; Servidio, 2014). As a result of the findings, it can be said that personal characteristics are effective in internet addiction, especially the effects of CONS, AGR and NEU characteristics is stronger.

These findings suggest that interventions tailored to personality traits could enhance the effectiveness of digital well-being programs. For example, students with low levels of conscientiousness may benefit from structured programs that promote time management, self-discipline, and goal setting through cognitive–behavioral techniques. Similarly, those scoring low on agreeableness may require support in developing empathy, interpersonal communication, and conflict resolution skills to reduce their reliance on online interactions for social support. In contrast, students high in neuroticism may particularly benefit from emotion regulation strategies, such as mindfulness-based stress reduction (MBSR) or dialectical behavior therapy (DBT), which aims to alleviate negative affect and impulsivity. Additionally, universal programs promoting self-regulation, emotional intelligence, and digital well-being skills across the student body may serve as non-stigmatizing preventive measures, benefiting even those without overt risk markers.

5.1. Limitations and future directions

Whilst the findings of this study highlight the efficacy of XAI methods in predicting IA, it is important to acknowledge the limitations of the study. Firstly, the sample was drawn from a single university in Türkiye, which may limit the generalizability of the findings to different socio-cultural contexts or age groups. Secondly, the study variables, including internet addiction, FoMO, loneliness, SIA, and personality traits, were measured using self-report scales. Nevertheless, it should be noted that self-report methods are subject to biases such as social desirability and response inaccuracies, which have the potential to affect the validity of the results. Thirdly, the data were collected cross-sectionally; therefore,

while associations between variables could be identified, it was not possible to establish causal relationships. It is recommended that future research employ longitudinal designs in order to facilitate a more comprehensive understanding of the development and interactions of IA over time. In conclusion, while a variety of machine learning models were employed in this study, the algorithms were constrained by predefined parameters (e.g., number of trees, depth limits), which may have restricted optimal model learning. The employment of alternative modelling techniques, deep learning methodologies, or other XAI approaches has the potential to further enhance the accuracy and interpretability of the results. Furthermore, although the random forest model demonstrated acceptable predictive performance ($R^2 = 0.59$ in training), a decline to 0.45 during 10-fold cross-validation suggests moderate overfitting. This performance drop may be attributed to model complexity, multicollinearity among predictors, or behavioral heterogeneity within the university student sample. Despite the use of stratified cross-validation to mitigate bias, these findings highlight the need for external validation using independent datasets. This moderate overfitting may be attributed to factors such as the high intercorrelation among psychological variables (e.g., FoMO, loneliness dimensions) and the relative homogeneity of the sample, which was drawn from a single institution. Employing feature selection or dimensionality reduction techniques in future models may help improve robustness and minimize reliance on sample-specific patterns. Future research should aim to test these models across diverse populations and consider longitudinal approaches to better assess their generalisability.

6. Conclusion

The present study makes a significant contribution to the extant literature by adopting an interdisciplinary approach to understanding IA, a common problem among university students, and by identifying key psychological and behavioural determinants using XAI. The findings emphasise that FoMO and SIA are the significant risk factors for IA, particularly among university students residing away from their families. Furthermore, the results suggest that social and emotional loneliness are closely associated with IA, with familial loneliness emerging as a particularly salient predictor. This highlights a potential link between emotional distance from one's family and excessive internet use as a coping mechanism among students. The study also reveals that personality traits play an important role in IA. Specifically, neuroticism was positively associated with higher addiction risk, while conscientiousness and agreeableness served as protective factors. A notable strength of this study lies in its methodological approach: in contradistinction to conventional statistical models, the application of machine learning and XAI techniques provided more nuanced and interpretable predictions of IA, offering deeper insight into the underlying psychological patterns.

While the study demonstrates the interpretability advantages of XAI in predicting IA, it is also essential to consider how students perceive and engage with such AI-based systems. As Wang et al. (2024) observed, students' acceptance of AI-driven interventions is often shaped more by familiarity and perceived usefulness than by technological novelty. Thus, to maximise the effectiveness of XAI models in educational and mental health contexts, their implementation should prioritise usability, transparency, and alignment with students' digital habits. Ensuring that these tools are perceived as supportive rather than intrusive may increase their adoption and impact in real-world settings. The integration of SHAP analysis, in particular, has the potential to enhance transparency in model interpretation, thereby providing actionable insights for mental health professionals, educators, and policymakers. In conclusion, the growing interest in XAI applications within the social sciences supports the development of more transparent and interpretable research methodologies (Elhai et al., 2020b; Miller, 2019). Future research should explore the longitudinal effects of psychological and behavioural predictors of IA, and XAI models should be validated across different cohorts to improve generalisability and inform tailored intervention strategies.

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Author contributions

Deniz Mertkan Gezgin: Conceptualization, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ebru Efeoğlu**: Formal analysis, Data curation, Software, Validation, Visualization, Writing – review & editing.

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Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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